

Amazon
 Microsoft
 MS
 Goldman Sachs
 Adobe
 Facebook
 :

Q Word Break

Given a dict (list of words [a-z]).

Given a string [a-z].

Check if the string can be broken down into valid words from the dictionary.

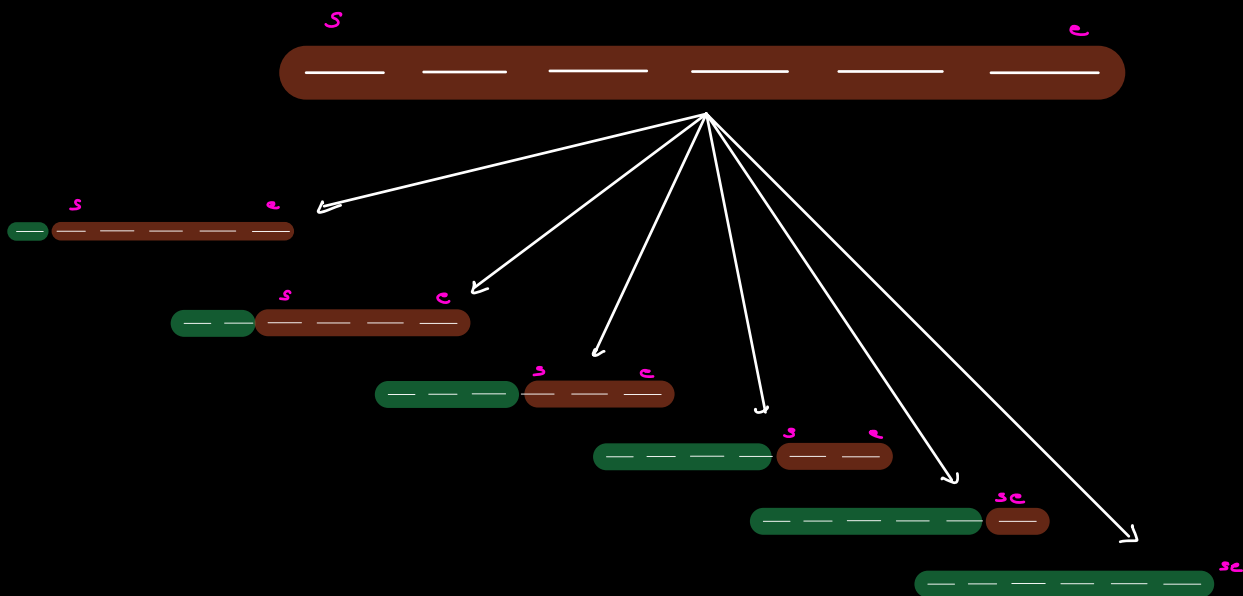
dict : { like, i, man, go, mangoes }

s : ilikemangoes ✓

s : ilikegoes ✗

s : ilikeimanmanmangoesgo ✓

s : mangoes ✓



wordBreak (A, s, ~~e~~) → Return true, if it is possible to break the substring from s to (N-1) {last index} into valid words from dict.

$\text{wordBreak}(A, 0, 5) \rightarrow$

check substr $A[0:i]$ in Set

$c \rightarrow A.length - 1$

$\text{isWord}(A, 0, 0) \ \&\& \ \text{wordBreak}(A, 1, 5)$
 $\parallel$
 $\text{isWord}(A, 0, 1) \ \&\& \ \text{wordBreak}(A, 2, 5)$
 $\parallel$
 $\text{isWord}(A, 0, 2) \ \&\& \ \text{wordBreak}(A, 3, 5)$
 $\parallel$
 $\text{isWord}(A, 0, 3) \ \&\& \ \text{wordBreak}(A, 4, 5)$
 $\parallel$
 $\text{isWord}(A, 0, 4) \ \&\& \ \text{wordBreak}(A, 5, 5)$
 $\parallel$
 $\text{isWord}(A, 0, 5) \ \&\& \ \text{wordBreak}(A, 6, 5)$

```

dict = { }
boolean wordBreak(A, s) {
    if (s == A.length()) { // clarify for ""
        ret true;
    }
    for (i = s;  $\rightarrow$  A.length()) {
        if (isWord(A, 0, i)
            &\& wordBreak(A, i+1)) {
            ret true;
        }
    }
    ret false;
}

```

TC (After Memoization)

m a n g o e s

dict : { like, i, man, go, mangoes }

m | angoes
ma | ngoes
man | goes

g | oes
go | es
e | s
es
goe | s
goes

Iterative

- Dimension 1D array
- $DP[i] \rightarrow T$ if str from i to N-1 can be broken down.
- Size man index $\rightarrow N$ (Base case)
size $\rightarrow N+1$

mang | oes
mango | es
mangoe | s
mangoes

S: i a m a b o y {am, a, boy, i, girl}
N = 7

DP:

0	1	2	3	4	5	6	7
T	T	F	T	T	F	F	T

y $\rightarrow$ isWord("y") $\rightarrow F$

o y
isWord("o") && WordBreak("y")
isWord("oy")

b o y

isWord("b")
isWord("bo")
isWord("boy") && WordBreak("")

3 4 5 6
a b o y

2 3 4 5 6
m a b o y

1 2 3 4 5 6
a m a b o y

0 1 2 3 4 5 6
i a m a b o y

isWord("a") & wordBreak("boy")

isWord("m") ✗
isWord("ma") ✗
isWord("mab") ✗
isWord("mabo") ✗
isWord("maboy") ✗ $\rightarrow O(N)$

} $O(N^2)$

isWord("a") & wordBreak("maboy")
isWord("am") & wordBreak("aboy")

isWord("i") & wordBreak("amaboy")

Q Palindrome Partitioning

Given a string.

Return, the min no of partitions required in the string such that every partition is a palindrome.

a a | b $\longrightarrow$ 1

a b b a $\longrightarrow$ 0

a | b a b b b a b | b | a b a b a $\longrightarrow$ 3

a b a | b | b b a b b | a b a b a

PalindromePartitions(A, s) = for (i = s $\longrightarrow$ A.length) {
 if (isPal(A, s, i)) {
 ans = min(ans, 1 + PalindromePartitions(A, i+1));
 }
}

└

Base Condition?

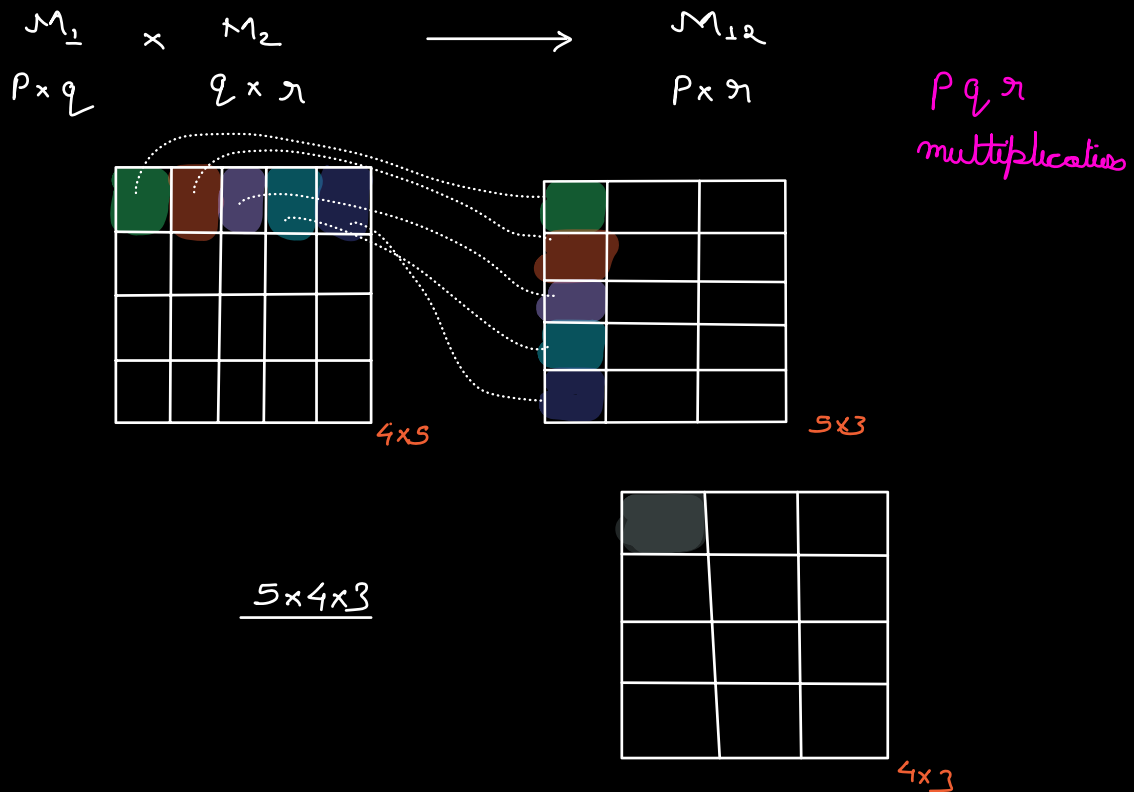
TC : $O(N^3)$ $\longrightarrow$ $O(N^2)$

Q. Matrix chain multiplication

Amazon
 Dribbble
 Samsung
 Mynt
 FB
 :

Given N matrix in a sequence such that it is possible to multiply them.

Return min no of multiplications required to multiply all the matrix.



$$M_1 \times M_2 \times M_3$$

$$(P \times Q) \quad (Q \times R) \quad (R \times S)$$

$$\begin{aligned}
 & (M_1 M_2) \times M_3 \\
 & \quad \downarrow \\
 & M_{12} \times M_3 \\
 & (P \times R) \quad (R \times S) \\
 & \quad \downarrow \\
 & P \times R \times S
 \end{aligned}$$

$PQR + PRS$

$$\begin{aligned}
 & M_1 \times (M_2 \times M_3) \\
 & \quad \downarrow \\
 & M_1 \times M_{23} \\
 & (P \times Q) \quad (Q \times S) \\
 & \quad \downarrow \\
 & P \times Q \times S
 \end{aligned}$$

$QRS + PQS$

p: 10

q: 30

r: 5

s: 60

$$pqr + prs$$

$$(10 \times 30 \times 5) + (10 \times 5 \times 60) \\ \Rightarrow 4500$$

$$qrs + prs$$

$$(30 \times 5 \times 60) + (10 \times 30 \times 60) \\ \Rightarrow 27,000$$

M₁

M₂

M₃

M₄

M₅

M₆

M₁

M₂

M₃

M₄

M₅

M₆

M₁

M₂

M₃

M₄

M₅

M₆

M₁

M₂

M₃

M₄

M₅

M₆

M₁

M₂

M₃

M₄

M₅

M₆

M₁

M₂

M₃

M₄

M₅

M₆

M₁

M₂

M₃

M₄

M₅

M₆

minMul(s, e) =

for (i = s → e) {

ans = min(ans, minMul(s, i)

+

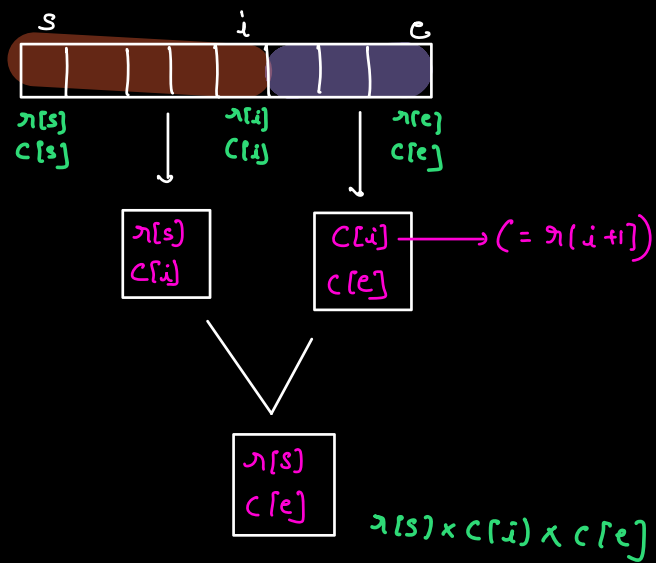
minMul(i+1, e)

+

a[s] * c[i] * c[e]

}

if (s == e) → 0



TC = ?

if / else
for / while
Array / String

Collections {
ArrayList
HashMap
HashSet
PQ
Queue
Stacks